

Question 1: Print Numbers from 1 to N

Source code:

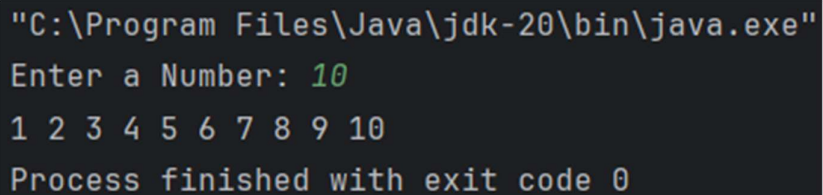
```
package LB_Assingment4;

import java.util.Scanner;

public class Numbers_1toN_Using_forloop {
    public static void main(String[] args) {
        Scanner sc=new Scanner(System.in);
        System.out.print("Enter a Number: ");
        int num=sc.nextInt();

        for(int i=1;i<=num;i++){
            System.out.print(i+" ");
        }
    }
}
```

Output:



```
"C:\Program Files\Java\jdk-20\bin\java.exe"
Enter a Number: 10
1 2 3 4 5 6 7 8 9 10
Process finished with exit code 0
```

Question 2: Print Multiples of 3 between 1 and N

Source code:

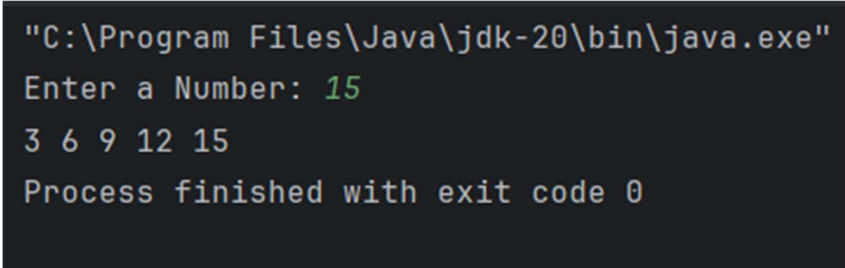
```
package LB_Assingment4;

import java.util.Scanner;

public class Mul_3_1toN_forloop {
    public static void main(String[] args) {
        Scanner sc=new Scanner(System.in);
        System.out.print("Enter a Number: ");
        int num=sc.nextInt();

        for(int i=1;i<=num;i++){
            int mul=3*i;
            if(mul<=num){
                System.out.print(mul+" ");
            }
        }
    }
}
```

Output:



```
"C:\Program Files\Java\jdk-20\bin\java.exe"
Enter a Number: 15
3 6 9 12 15
Process finished with exit code 0
```

Question 3: Calculate the Factorial of a Number

Source code:

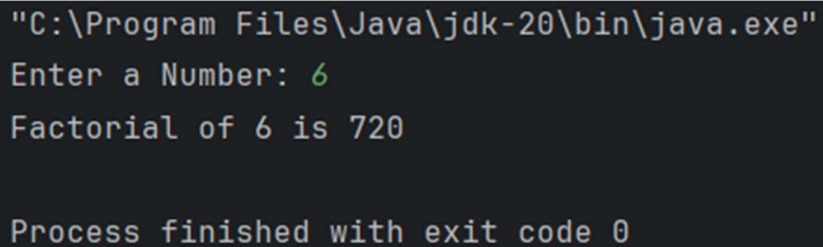
```
package LB_Assingment4;

import java.util.Scanner;

public class Fact_of_num_forloop {
    public static void main(String[] args) {
        Scanner sc=new Scanner(System.in);
        System.out.print("Enter a Number: ");
        int num=sc.nextInt();

        int mul=1;
        for(int i=1;i<=num;i++){
            mul=mul*i;
        }
        System.out.println("Factorial of "+num+" is "+mul);
    }
}
```

Output:

A screenshot of a terminal window showing the execution of a Java program. The command prompt is "C:\Program Files\Java\jdk-20\bin\java.exe". The program prompts "Enter a Number:" and the user enters "6". The program outputs "Factorial of 6 is 720". The terminal also shows "Process finished with exit code 0".

```
"C:\Program Files\Java\jdk-20\bin\java.exe"
Enter a Number: 6
Factorial of 6 is 720

Process finished with exit code 0
```

Question 4: Print Even Numbers from 1 to N

Source code:

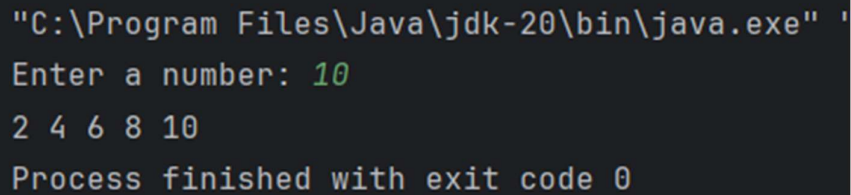
```
package LB_Assingment4;

import java.util.Scanner;

public class Even_no_forloop {
    public static void main(String[] args) {
        Scanner sc=new Scanner(System.in);
        System.out.print("Enter a number: ");
        int num=sc.nextInt();

        for(int i=1;i<=num;i++){
            if(i%2==0){
                System.out.print(i+" ");
            }
        }
    }
}
```

Output:



```
"C:\Program Files\Java\jdk-20\bin\java.exe" '
Enter a number: 10
2 4 6 8 10
Process finished with exit code 0
```

Question 5: Sum of Odd Numbers between 1 and N

Source code:

```
package LB_Assingment4;

import java.util.Scanner;

public class Odd_no_forloop {
    public static void main(String[] args) {
        Scanner sc=new Scanner(System.in);
        System.out.print("Enter a Number: ");
        int num=sc.nextInt();
        int add=0;
        for(int i=1;i<=num;i++){

            if(i%2!=0){
                add=add+i;
            }
        }
        System.out.println("The Sum of odd numbers from 1 to "+num+" is: "+add);
    }
}
```

Output:

```
"C:\Program Files\Java\jdk-20\bin\java.exe"
Enter a Number: 10
The Sum of odd numbers from 1 to 10 is: 25

Process finished with exit code 0
```

Question 6: Print All Elements of an Array

Source code:

```
package LB_Assingment4;

import java.util.Scanner;

public class Print_array_elements {
    public static void main(String[] args) {
        Scanner sc=new Scanner(System.in);
        int [] arr=new int[5];
        System.out.print("Enter 5 Intergers: ");

        for(int i=0;i<arr.length;i++){
            arr[i]= sc.nextInt();
        }
        for(int num:arr){
            System.out.print(num+" ");
        }
    }
}
```

Output:

```
"C:\Program Files\Java\jdk-20\bin\java.exe"
Enter 5 Intergers: 10 50 4 6 20
10 50 4 6 20
Process finished with exit code 0
|
```

Question 7: Find the Sum of All Elements in an Array

Source code:

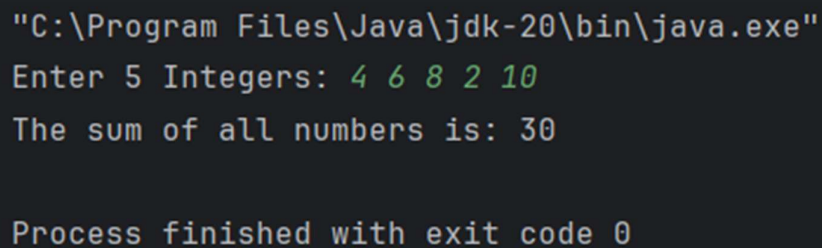
```
package LB_Assingment4;

import java.util.Scanner;

public class sum_of_arr_elements_foreach {
    public static void main(String[] args) {
        Scanner sc=new Scanner(System.in);
        int []arr=new int[5];

        System.out.print("Enter 5 Integers: ");
        for(int i=0;i<arr.length;i++){
            arr[i]= sc.nextInt();
        }
        int add=0;
        for(int num:arr){
            add=add+num;
        }
        System.out.println("The sum of all numbers is: "+add);
    }
}
```

Output:

A screenshot of a Windows command prompt window showing the execution of a Java program. The path to the Java executable is displayed at the top. The program prompts the user to enter 5 integers, which are entered as 4, 6, 8, 2, and 10. The program then outputs the sum of these numbers, which is 30. Finally, it shows the process finished with exit code 0.

```
"C:\Program Files\Java\jdk-20\bin\java.exe"
Enter 5 Integers: 4 6 8 2 10
The sum of all numbers is: 30

Process finished with exit code 0
```

Question 8: Print All Names in a String Array

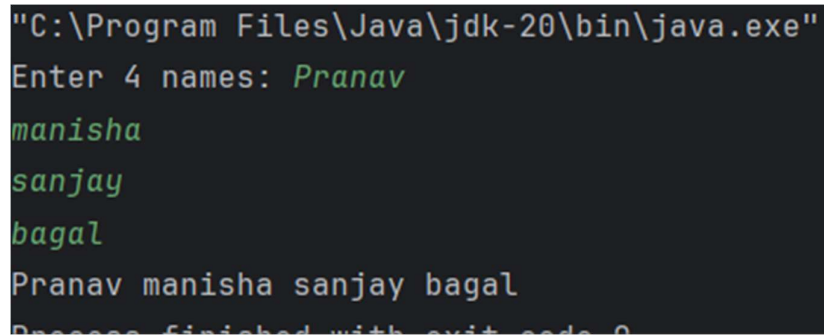
Source code:

```
package LB_Assingment4;

import java.util.Scanner;

public class print_names_array {
    public static void main(String[] args) {
        Scanner sc=new Scanner(System.in);
        String [] arr=new String[4];
        System.out.println("Enter 4 names: ");
        for(int i=0;i<arr.length;i++){
            arr[i]=sc.nextLine();
        }
        for(String names:arr){
            System.out.print(names+" ");
        }
    }
}
```

Output:

A screenshot of a terminal window showing the execution of a Java program. The first line is the command prompt: "C:\Program Files\Java\jdk-20\bin\java.exe". The program prompts "Enter 4 names:" and the user enters "Pranav" on the next line, "manisha" on the third line, "sanjay" on the fourth line, and "bagal" on the fifth line. The sixth line shows the output: "Pranav manisha sanjay bagal". The seventh line shows the program finishing with "Process finished with exit code 0".

```
"C:\Program Files\Java\jdk-20\bin\java.exe"
Enter 4 names: Pranav
manisha
sanjay
bagal
Pranav manisha sanjay bagal
Process finished with exit code 0
```


Question 9: Find the Largest Element in an Array

Source code:

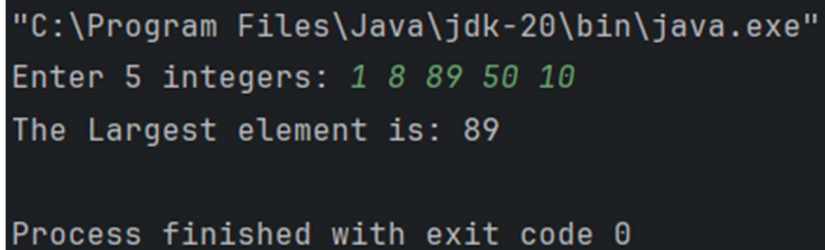
```
package LB_Assingment4;

import java.util.Arrays;
import java.util.Scanner;

public class largest_num_arr {
    public static void main(String[] args) {
        Scanner sc=new Scanner(System.in);
        int []arr=new int[5];
        System.out.print("Enter 5 integers: ");

        for(int i=0;i< arr.length;i++){
            arr[i]=sc.nextInt();
        }
        Arrays.sort(arr);
        System.out.println("The Largest element is: "+arr[arr.length-1]);
    }
}
```

Output:



```
"C:\Program Files\Java\jdk-20\bin\java.exe"
Enter 5 integers: 1 8 89 50 10
The Largest element is: 89

Process finished with exit code 0
```

Question 10: Find the Average of Elements in an Array

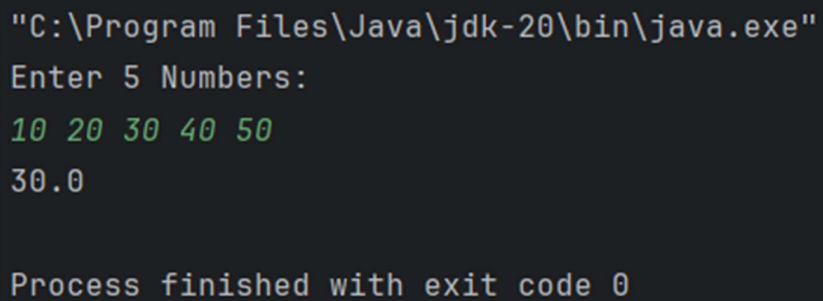
Source code:

```
package LB_Assingment4;

import java.util.Scanner;

public class Avg_arr_elements {
    public static void main(String[] args) {
        Scanner sc=new Scanner(System.in);
        int [] arr=new int[5];
        System.out.println("Enter 5 Numbers: ");
        for(int i=0;i< arr.length;i++){
            arr[i]=sc.nextInt();
        }
        int sum=0;
        for(int i=0;i<arr.length;i++){
            sum=sum+arr[i];
        }
        double avg=sum/ arr.length;
        System.out.println(avg);
    }
}
```

Output:



```
"C:\Program Files\Java\jdk-20\bin\java.exe"
Enter 5 Numbers:
10 20 30 40 50
30.0

Process finished with exit code 0
```

Question 11: Count Positive and Negative Numbers in an Array

Source code:

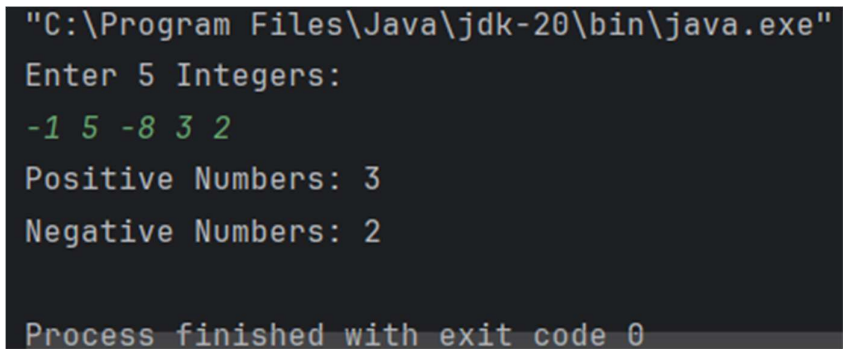
```
package LB_Assingment4;

import java.util.Scanner;

public class Count_positive_Negative_nos {
    public static void main(String[] args) {
        Scanner sc=new Scanner(System.in);
        int [] arr=new int[5];
        System.out.println("Enter 5 Integers: ");

        for(int i=0;i< arr.length;i++){
            arr[i]=sc.nextInt();
        }
        int count_pos=0;
        int count_neg=0;
        for(int i=0;i< arr.length;i++){
            if(arr[i]>0){
                count_pos=count_pos+1;
            }else{
                count_neg=count_neg+1;
            }
        }
        System.out.println("Positive Numbers: "+count_pos);
        System.out.println("Negative Numbers: "+count_neg);
    }
}
```

Output:



The screenshot shows a terminal window with the following text:

```
"C:\Program Files\Java\jdk-20\bin\java.exe"
Enter 5 Integers:
-1 5 -8 3 2
Positive Numbers: 3
Negative Numbers: 2
Process finished with exit code 0
```

The input values are -1, 5, -8, 3, and 2. The program correctly identifies 3 positive numbers (5, 3, 2) and 2 negative numbers (-1, -8).

Question 12: Sort an Array in Ascending Order

Source code:

```
package LB_Assingment4;

import java.util.Arrays;
import java.util.Scanner;

public class Array_sorting {
    public static void main(String[] args) {
        Scanner sc=new Scanner(System.in);
        int [] arr=new int[5];
        System.out.println("Enter 5 Integers: ");

        for(int i=0;i< arr.length;i++){
            arr[i]=sc.nextInt();
        }
        Arrays.sort(arr);
        System.out.print("Sorted Array: ");
        for(int num:arr){
            System.out.print(num+" ");
        }
    }
}
```

Output:

```
"C:\Program Files\Java\jdk-20\bin\java.exe"
Enter 5 Integers:
5 3 4 1 2
Sorted Array: 1 2 3 4 5
Process finished with exit code 0
```

Question 13: Check if an Array Contains a Specific Element

Source code:

```
package LB_Assingment4;

import java.util.Arrays;
import java.util.Scanner;

public class Check_arr_contain_elements {
    public static void main(String[] args) {
        Scanner sc=new Scanner(System.in);
        Integer [] arr=new Integer[5];
        System.out.print("Enter 5 Integers: ");

        for(int i=0;i< arr.length;i++){
            arr[i]=sc.nextInt();
        }
        System.out.print("Enter the Number to Search: ");
        int find_num=sc.nextInt();
        if(Arrays.asList(arr).contains(find_num)){
            System.out.println("Found");
        }else{
            System.out.println("Not Found");
        }
    }
}
```

Output:

```
"C:\Program Files\Java\jdk-20\bin\java.exe"
Enter 5 Integers: 10 20 30 40 50
Enter the Number to Search: 40
Found

Process finished with exit code 0
```

Question 14: Find the Index of an Element in an Array

Source code:

```
package LB_Assingment4;

import java.util.Arrays;
import java.util.Scanner;

public class Index_of_elem_arr {
    public static void main(String[] args) {
        Scanner sc=new Scanner(System.in);
        int [] arr=new int[5];
        System.out.print("Enter 5 Integers: ");
        for(int i=0;i<arr.length;i++){
            arr[i]= sc.nextInt();
        }
        System.out.print("Enter the Number to search: ");
        int find_num= sc.nextInt();

        int index=Arrays.binarySearch(arr,find_num);
        if(index>=0){
            System.out.println("The number "+find_num+" is found at index "+index);
        }else{
            System.out.println("Not found");
        }
    }
}
```

Output:

```
"C:\Program Files\Java\jdk-20\bin\java.exe"
Enter 5 Integers: 5 10 15 20 25
Enter the Number to search: 15
The number 15 is found at index 2

Process finished with exit code 0
```

Question 15:Write a program to print the following pattern:

```
1
2*2
3*3*3
4*4*4*4
5*5*5*5*5
5*5*5*5*5
4*4*4*4
3*3*3
2*2
```

Source code:

```
package LB_Assingment4;

public class Q15 {
    public static void main(String[] args) {
        for(int i=1;i<=5;i++){
            for(int j=1;j<=i;j++){
                System.out.print(i);
                if(j<i){
                    System.out.print("*");
                }
            }
            System.out.println();
        }
        for(int i=5;i>=2;i--){
            for(int j=1;j<=i;j++){
                System.out.print(i);
                if(j<i){
                    System.out.print("*");
                }
            }
            System.out.println();
        }
    }
}
```

Output:

```
"C:\Program Files\Java\jdk-20\bin\java.exe"
1
2*2
3*3*3
4*4*4*4
5*5*5*5*5
5*5*5*5*5
4*4*4*4
3*3*3
2*2
```

Question 16: Write a program to print the following pattern:

```
1
1*2
1*2*3
1*2*3*4
1*2*3*4*5
```

Source code:

```
package LB_Assingment4;

public class Q16 {
    public static void main(String[] args) {
        for(int i=1;i<=5;i++){
            for(int j=1;j<=i;j++){
                System.out.print(j);
                if(j<i){
                    System.out.print("*");
                }
            }
            System.out.println();
        }
    }
}
```

Output:

```
"C:\Program Files\Java\jdk-20\bin\java.exe"
1
1*2
1*2*3
1*2*3*4
1*2*3*4*5

Process finished with exit code 0
```


Question 17: Write a program to print the following pattern:

```
1
1*3
1*3*5
1*3*5*7
1*3*5*7*9
```

Source code:

```
package LB_Assingment4;

public class Q17 {
    public static void main(String[] args) {
        for(int i=1;i<=5;i++){
            for(int j=1;j<=2*i-1;j++){
                if(j%2!=0){
                    System.out.print(j);

                    if(j<2*i-1){
                        System.out.print("*");
                    }
                }
            }
            System.out.println();
        }
    }
}
```

Output:

```
"C:\Program Files\Java\jdk-20\bin\java.exe"
1
1*3
1*3*5
1*3*5*7
1*3*5*7*9

Process finished with exit code 0
```

Question 18: Write a program to print the following pattern:

```
11111
22222
33333
44444
55555
```

Source code:

```
package LB_Assingment4;

public class Q18 {
    public static void main(String[] args) {
        for(int i=1;i<=5;i++){
            for(int j=1;j<=5;j++){
                System.out.print(i);
            }
            System.out.println();
        }
    }
}
```

Output:

```
"C:\Program Files\Java\jdk-20\bin\java.exe" "-
11111
22222
33333
44444
55555

Process finished with exit code 0
```

Question 19: Write a program to print the following pattern:

```
1
22
333
4444
55555
```

Source code:

```
package LB_Assingment4;

public class Q19 {
    public static void main(String[] args) {
        for(int i=1;i<=5;i++){
            for(int j=1;j<=i;j++){
                System.out.print(i);
            }
            System.out.println();
        }
    }
}
```

Output:

```
"C:\Program Files\Java\jdk-20\bin\java.exe"
1
22
333
4444
55555

Process finished with exit code 0
```

Question 20: Write a program to print the following pattern

```
1
12
123
1234
12345
```

Source code:

```
package LB_Assingment4;

public class Q20 {
    public static void main(String[] args) {
        for(int i=1;i<=5;i++){
            for(int j=1;j<=i;j++){
                System.out.print(j);
            }
            System.out.println();
        }
    }
}
```

Output:

```
"C:\Program Files\Java\jdk-20\bin\java.exe"
1
12
123
1234
12345

Process finished with exit code 0
```

Question 21: Write a program to print the following pattern:

```
1
2 3
4 5 6
7 8 9 10
11 12 13 14 15
```

Source code:

```
package LB_Assingment4;

public class Q21 {
    public static void main(String[] args) {
        int num=1;
        for(int i=1;i<=5;i++){
            for(int j=1;j<=i;j++){
                System.out.print(num+" ");
                num++;
            }
            System.out.println();
        }
    }
}
```

Output:

```
"C:\Program Files\Java\jdk-20\bin\java.exe"
1
2 3
4 5 6
7 8 9 10
11 12 13 14 15

Process finished with exit code 0
```

Question 22: Write a program to print the following pattern:

```
*****
*       *
*       *
*       *
*       *
*****
```

Source code:

```
package LB_Assingment4;

public class Q22 {
    public static void main(String[] args) {
        for(int i=1;i<=6;i++){
            for(int j=1;j<=6;j++){
                if(i==1 || i==6 || j==1 || j==6){
                    System.out.print("*");
                }else{
                    System.out.print(" ");
                }
            }
            System.out.println();
        }
    }
}
```

Output:

```
"C:\Program Files\Java\jdk-20\bin\java.exe"
*****
*       *
*       *
*       *
*       *
*****

Process finished with exit code 0
```

Question 23: Write a program to print the following pattern:



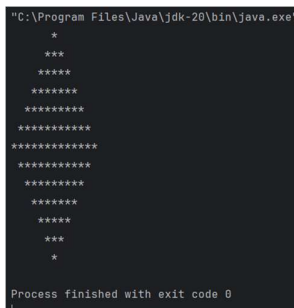
```
*
***
*****
*****
*****
*****
*****
*****
***
*
```

Source code:

```
package LB_Assingment4;

public class Q23 {
    public static void main(String[] args) {
        for(int i=1;i<=7;i++){
            for(int j=1;j<=7-i;j++){
                System.out.print(" ");
            }
            for(int j=1;j<=2*i-1;j++){
                System.out.print("*");
            }
            System.out.println();
        }
        for(int i=6;i>=1;i--){
            for(int j=1;j<=7-i;j++){
                System.out.print(" ");
            }
            for(int j=1;j<=2*i-1;j++){
                System.out.print("*");
            }
            System.out.println();
        }
    }
}
```

Output:



```
"C:\Program Files\Java\jdk-20\bin\java.exe"
*
***
*****
*****
*****
*****
*****
*****
***
*
Process finished with exit code 0
```

Question 24: Reverse a String

Source code:

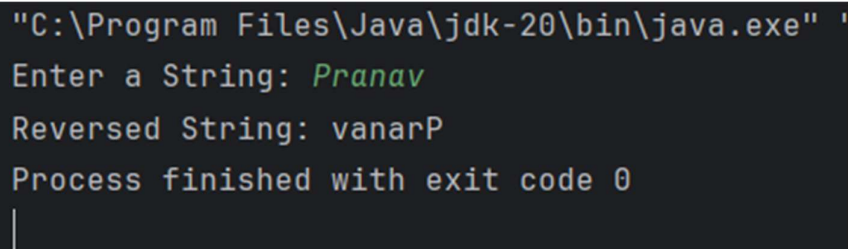
```
package LB_Assingment4;

import java.util.Scanner;

public class Q24 {
    public static void main(String[] args) {
        Scanner sc=new Scanner(System.in);
        System.out.print("Enter a String: ");
        String word=sc.nextLine();

        System.out.print("Reversed String: ");
        for(int i=word.length()-1;i>=0;i--){
            System.out.print(word.charAt(i));
        }
    }
}
```

Output:



```
"C:\Program Files\Java\jdk-20\bin\java.exe" '
Enter a String: Pranav
Reversed String: vanarP
Process finished with exit code 0
|
```


Question 25: Count Vowels in a String

Source code:

```
package LB_Assingment4;

import java.util.Scanner;

public class Q25 {
    public static void main(String[] args) {
        Scanner sc=new Scanner(System.in);
        System.out.print("Enter a String: ");
        String word=sc.nextLine();

        int count=0;
        for(int i=0;i<word.length();i++){
            char ch=word.charAt(i);
            if(ch == 'a' ||ch == 'e' ||ch == 'i' ||ch == 'o' ||ch == 'u'
                ||ch == 'A' ||ch == 'E' ||ch == 'I' ||ch == 'O' ||ch == 'U'){
                count++;
            }
        }
        System.out.println("The number of vowels in '" + word + "' is: " + count);
    }
}
```

Output:

```
"C:\Program Files\Java\jdk-20\bin\java.exe"
Enter a String: Rainbow
The number of vowels in 'Rainbow' is: 3

Process finished with exit code 0
```

Question 26: Check if a String is a Palindrome

Source code:

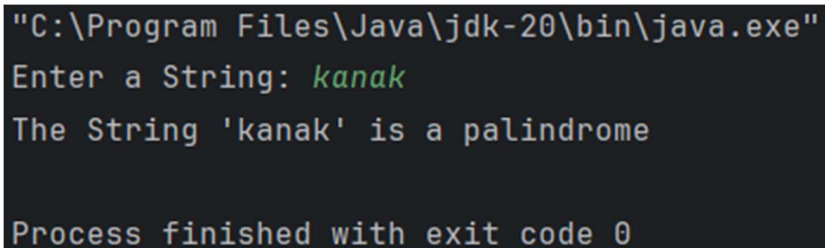
```
package LB_Assingment4;

import java.util.Scanner;

public class Q26 {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter a String: ");
        String word = sc.nextLine();

        String rev = "";
        for (int i=word.length()-1;i>=0;i--){
            rev =rev+word.charAt(i);
        }
        if(word.equals(rev)){
            System.out.println("The String '"+word+"' is a palindrome");
        }else{
            System.out.println("The String '"+word+"' is not a palindrome");
        }
    }
}
```

Output:



```
"C:\Program Files\Java\jdk-20\bin\java.exe"
Enter a String: kanak
The String 'kanak' is a palindrome

Process finished with exit code 0
```

Question 27: String Literal and Object Creation

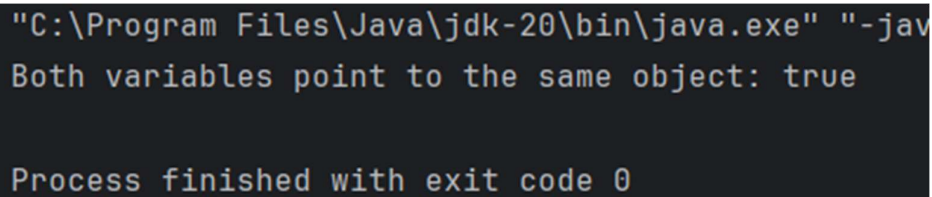
Source code:

```
package LB_Assingment4;

public class Q27 {
    public static void main(String[] args) {
        String str1="hello";
        String str2="hello";

        if(str1==str2){
            System.out.println("Both variables point to the same object: "+(str1==str2));
        }else {
            System.out.println("Both variables point to the same object: "+(str1==str2));
        }
    }
}
```

Output:



```
"C:\Program Files\Java\jdk-20\bin\java.exe" "-jav
Both variables point to the same object: true

Process finished with exit code 0
```

Question 28: String Creation with new Keyword

Source code:

```
package LB_Assingment4;

public class Q28 {
    public static void main(String[] args) {
        String str1=new String("Hello");
        String str2=new String("Hello");

        System.out.println("Using ==: "+(str1==str2));
        System.out.println("Using .equals(): "+(str1.equals(str2)));
    }
}
```

Output:

```
"C:\Program Files\Java\jdk-20\bin\java.exe"
Using ==: false
Using .equals(): true

Process finished with exit code 0
```

Question 29: String Concatenation and Object Creation

Source code:

```
package LB_Assingment4;

public class Q29 {
    public static void main(String[] args) {
        String str1="Hello";
        String str2="World";
        String str3=str1+str2;

        System.out.println("Is str3 pointing to the same object as str1? "+(str1==str3));
    }
}
```

Output:

```
"C:\Program Files\Java\jdk-20\bin\java.exe" "-javaagent:C
Is str3 pointing to the same object as str1? false

Process finished with exit code 0
```

Question 30: String Pool with intern() Method

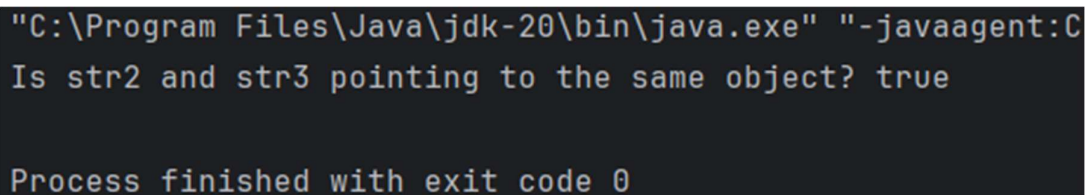
Source code:

```
package LB_Assingment4;

public class Q30 {
    public static void main(String[] args) {
        String str1=new String("Hello");
        String str2=str1.intern();
        String str3="Hello";

        System.out.println("Is str2 and str3 pointing to the same object? "+(str2==str3));
    }
}
```

Output:

A screenshot of a Java command prompt window. The first line shows the command: "C:\Program Files\Java\jdk-20\bin\java.exe" "-javaagent:C:\P. The second line shows the output: Is str2 and str3 pointing to the same object? true. The third line shows the exit message: Process finished with exit code 0.

```
"C:\Program Files\Java\jdk-20\bin\java.exe" "-javaagent:C:\P
Is str2 and str3 pointing to the same object? true

Process finished with exit code 0
```

Question 31: Multiple String Literals with Same Content

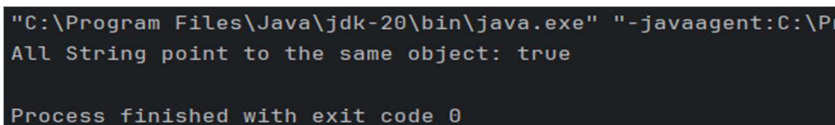
Source code:

```
package LB_Assingment4;

public class Q31 {
    public static void main(String[] args) {
        String str1="Java";
        String str2="Java";
        String str3="Java";

        System.out.println("All String point to the same object: "+(str1==str2 && str2==str3));
    }
}
```

Output:

A screenshot of a Java command prompt window. The first line shows the command: "C:\Program Files\Java\jdk-20\bin\java.exe" "-javaagent:C:\P. The second line shows the output: All String point to the same object: true. The third line shows the exit message: Process finished with exit code 0.

```
"C:\Program Files\Java\jdk-20\bin\java.exe" "-javaagent:C:\P
All String point to the same object: true

Process finished with exit code 0
```